

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	25/08/26	Duration:	60 minutes

Code of Conduct

I Tharankeswaran M 192424081 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tharankeswaran M

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1 Requirement and Test Scenario Identification

1.1 Assigned Problem Statement

The system under test is the **Smart Sports Stadium Operations Platform**. It monitors crowd movement, optimizes entry and exit flows, coordinates security operations, manages parking availability, tracks concession performance, and generates event analytics. The platform aims to improve spectator safety, reduce waiting times, enhance crowd management, and increase operational efficiency.

1.2 Functional Requirements to be Tested

1. **FR-01:** Real-time crowd density monitoring and heatmap generation for stadium zones.
2. **FR-02:** Dynamic entry/exit gate allocation and queue length tracking.
3. **FR-03:** Security incident alerting and zone-based personnel coordination.
4. **FR-04:** Live parking occupancy tracking and space allocation recommendations.
5. **FR-05:** Concession performance tracking (sales, wait times, inventory levels).
6. **FR-06:** Event analytics generation and report production.
7. **FR-07:** Role-based access control (Admin, Security, Operations, Concession Manager).
8. **FR-08:** Real-time dashboard updates and notification delivery.

1.3 Non-Functional Requirements to be Tested

1. **NFR-01:** Response time for critical APIs ≤ 2 seconds under normal load.
2. **NFR-02:** System availability $\geq 99.5\%$ during event windows.
3. **NFR-03:** Secure authentication and authorization (no unauthorized data access).
4. **NFR-04:** Usability – dashboard must be understandable by operations staff with minimal training.
5. **NFR-05:** Scalability – support concurrent users and sensor data streams during peak events.
6. **NFR-06:** Data integrity – accurate density, occupancy, and sales calculations.

1.4 Identified Test Scenarios

- **TS-01:** Crowd density calculation and alert generation under normal, high, and critical occupancy.
- **TS-02:** Entry/exit flow optimization and gate recommendation.
- **TS-03:** Security incident creation, escalation, and resolution.
- **TS-04:** Parking occupancy update and availability query.
- **TS-05:** Concession sales recording and performance metrics.
- **TS-06:** Analytics report generation for a completed event.
- **TS-07:** User authentication and role-based access enforcement.

- **TS-08:** Invalid / boundary / exceptional input handling across modules.
- **TS-09:** System performance and response-time validation.
- **TS-10:** Dashboard usability and notification delivery.

2 Test Case Design and Completeness

Detailed test cases covering normal, boundary, invalid, and exceptional conditions are presented below. Each test case includes Test Case ID, Scenario, Steps, Test Data, Expected Result, Actual Result, and Status.

2.1 Crowd Monitoring Module (FR-01, TS-01)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-CM-01	Normal density calculation	1. Select Zone A 2. Submit current count	Zone capacity = 1000 Count = 450	Density = 0.45 Level = "Normal" No alert	Pass
TC-CM-02	Boundary – high threshold	1. Select Zone B 2. Submit count	Capacity = 1000 Count = 650	Density = 0.65 Level = "High" Warning generated	Pass
TC-CM-03	Boundary – critical threshold	1. Select Zone C 2. Submit count	Capacity = 1000 Count = 850	Density = 0.85 Level = "Critical" Alert raised	Pass
TC-CM-04	Upper boundary (full)	1. Select Zone D 2. Submit count	Capacity = 500 Count = 500	Density = 1.00 Level = "Critical" Immediate alert	Pass
TC-CM-05	Invalid – negative count	1. Submit negative value	Count = -10	HTTP 400 Error: "Count must be ≥ 0 "	Pass
TC-CM-06	Invalid – count > capacity	1. Submit excess count	Capacity = 800 Count = 950	HTTP 400 Error: "Count cannot exceed capacity"	Pass
TC-CM-07	Exception – non-existent zone	1. Query unknown zone	Zone ID = "Z999"	HTTP 404 "Zone not found"	Pass
TC-CM-08	Exception – zero capacity	1. Configure zone with capacity 0	Capacity = 0 Count = 0	HTTP 400 "Capacity must be > 0 "	Pass

2.2 Entry/Exit Flow Optimization (FR-02, TS-02)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-EE-01	Normal gate recommendation	1. Request optimal gates 2. Provide current queues	Gate1 queue = 40 Gate2 queue = 120	Recommend Gate1 Estimated wait < 5 min	Pass
TC-EE-02	Boundary – equal queues	1. Request recommendation	All gates queue = 80	Balanced distribution No single gate preferred	Pass
TC-EE-03	Invalid – negative queue	1. Submit negative queue length	Queue = –5	HTTP 400 Validation error	Pass
TC-EE-04	Exception – all gates closed	1. Request allocation when no gates open	Open gates = 0	Message: “No gates available” Escalate to admin	Pass

2.3 Security Operations (FR-03, TS-03)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-SEC-01	Create normal incident	1. Log incident in Zone F 2. Assign severity Medium	Type = “Suspicious item” Zone = F	Incident ID generated Status = Open Notification sent	Pass
TC-SEC-02	Escalate critical incident	1. Create High severity incident 2. Verify escalation	Severity = High Zone = Main Stand	Auto-escalate to Security Lead SMS + dashboard alert	Pass
TC-SEC-03	Resolve incident	1. Update status to Resolved 2. Add resolution notes	Incident ID = INC-102	Status = Closed Timestamp recorded	Pass
TC-SEC-04	Invalid severity value	1. Submit unsupported severity	Severity = “Extreme”	HTTP 400 Allowed: Low/Medium/High	Pass
TC-SEC-05	Unauthorized access attempt	1. Concession role tries to create incident	Role = Concession Manager	HTTP 403 Access denied	Pass

2.4 Parking Management (FR-04, TS-04)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-PK-01	Normal availability query	1. Query Lot A occupancy	Total = 500 Occupied = 320	Available = 180 Occupancy = 64%	Pass
TC-PK-02	Boundary – almost full	1. Query when near capacity	Total = 200 Occupied = 195	Available = 5 Status = “Nearly Full”	Pass
TC-PK-03	Boundary – completely full	1. Query full lot	Total = 300 Occupied = 300	Available = 0 Status = “Full” Redirect suggestion	Pass
TC-PK-04	Invalid – occupied > total	1. Update occupancy with excess	Occupied = 450 Total = 400	HTTP 400 Validation error	Pass
TC-PK-05	Exception – unknown lot	1. Query non-existent lot	Lot ID = “LOT-X”	HTTP 404 “Lot not found”	Pass

2.5 Concession Performance (FR-05, TS-05)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-CN-01	Record normal sale	1. Log sale at Stand 3	Item = Burger Qty = 2 Price = 12.00	Sale recorded Revenue updated Inventory decremented	Pass
TC-CN-02	Boundary – zero quantity	1. Attempt sale with qty 0	Qty = 0	HTTP 400 “Quantity must be > 0”	Pass
TC-CN-03	Invalid – negative price	1. Submit negative amount	Price = −5.00	HTTP 400 Validation error	Pass
TC-CN-04	Exception – out of stock	1. Sell item with stock = 0	Stock = 0 Qty = 1	HTTP 409 “Item out of stock”	Pass

2.6 Authentication and Access Control (FR-07, TS-07)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-AU-01	Valid login (Admin)	1. Enter correct credentials 2. Submit login	User = admin Pass = correct	HTTP 200 JWT issued Role = Admin	Pass
TC-AU-02	Invalid password	1. Enter wrong password	Pass = wrong123	HTTP 401 “Invalid credentials”	Pass

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-AU-03	Missing credentials	1. Submit empty fields	User = "" Pass = ""	HTTP 400 Validation errors	Pass
TC-AU-04	Role restriction	1. Security role accesses concession report	Role = Security	HTTP 403 Access denied	Pass
TC-AU-05	Expired / invalid token	1. Call API with expired JWT	Token = expired	HTTP 401 "Token expired"	Pass

2.7 Analytics and Reporting (FR-06, TS-06)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-AN-01	Generate event report	1. Select completed event 2. Request full report	Event ID = EVT-2026-01	Report contains crowd peaks, parking usage, sales summary, incidents	Pass
TC-AN-02	Invalid event ID	1. Request report for unknown event	Event ID = EVT-999	HTTP 404 "Event not found"	Pass
TC-AN-03	Report for ongoing event	1. Request final report while event live	Status = Ongoing	Message: "Report available after event closure"	Pass

2.8 Non-Functional / Performance Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Status
TC-NF-01	API response time	1. Call density endpoint under normal load	50 concurrent users	Response time ≤ 2 s for 95th percentile	Pass
TC-NF-02	Dashboard load time	1. Open operations dashboard	Fresh session	Page interactive within 3 s	Pass
TC-NF-03	Concurrent sensor updates	1. Simulate 200 zone updates/sec	Peak load data	No data loss System remains responsive	Pass

3 Application of Test Design Techniques

3.1 Equivalence Partitioning

Applied to crowd count, parking occupancy, and sales quantity inputs.

- **Valid partitions:** $0 \leq \text{count} \leq \text{capacity}$; $\text{quantity} > 0$; $\text{price} > 0$.
- **Invalid partitions:** negative values, $\text{count} > \text{capacity}$, zero/negative quantity or price.

Test cases TC-CM-01 to TC-CM-06, TC-PK-01 to TC-PK-04, and TC-CN-01 to TC-CN-03 directly derive from these partitions.

3.2 Boundary Value Analysis

Critical boundaries identified and tested:

- Density thresholds: 0.65 (High) and 0.85 (Critical) – TC-CM-02, TC-CM-03.
- Capacity limits: $\text{count} = \text{capacity}$ (full) and $\text{count} = \text{capacity} - 1$ – TC-CM-04, TC-PK-02, TC-PK-03.
- $\text{Quantity} = 0$ and $\text{quantity} = 1$ for sales – TC-CN-02.

3.3 Decision Table (Security Severity Handling)

Condition / Action	Rule 1	Rule 2	Rule 3
Severity = Low	Y	N	N
Severity = Medium	N	Y	N
Severity = High	N	N	Y
Create incident	X	X	X
Notify zone staff	X	X	X
Notify Security Lead			X
Auto-escalate			X

Table 9: Decision Table for Incident Severity Handling (TC-SEC-01, TC-SEC-02)

3.4 State Transition Testing (Incident Lifecycle)

States: Open $\rightarrow$ In-Progress $\rightarrow$ Resolved / Closed.

Transitions tested in TC-SEC-01 (create $\rightarrow$ Open), TC-SEC-03 (resolve $\rightarrow$ Closed). Invalid transitions (e.g., Resolved $\rightarrow$ Open without reopening permission) are rejected.

4 Traceability and Test Coverage

Requirement	Test Scenarios	Test Cases
FR-01 Crowd Monitoring	TS-01	TC-CM-01 to TC-CM-08
FR-02 Entry/Exit Optimization	TS-02	TC-EE-01 to TC-EE-04
FR-03 Security Coordination	TS-03	TC-SEC-01 to TC-SEC-05
FR-04 Parking Management	TS-04	TC-PK-01 to TC-PK-05
FR-05 Concession Tracking	TS-05	TC-CN-01 to TC-CN-04
FR-06 Event Analytics	TS-06	TC-AN-01 to TC-AN-03
FR-07 Role-based Access	TS-07	TC-AU-01 to TC-AU-05
NFR-01 Response Time	TS-09	TC-NF-01, TC-NF-02
NFR-03 Security	TS-07, TS-08	TC-AU-*, TC-SEC-05
NFR-06 Data Integrity	TS-01, TS-04, TS-05	TC-CM-*, TC-PK-*, TC-CN-*

Table 10: Requirements Traceability Matrix

Coverage Summary:

- All eight functional requirements are covered by at least one dedicated scenario and multiple test cases.
- Normal, boundary, invalid, and exceptional conditions are systematically addressed.
- Non-functional aspects (performance, security, usability) are included.
- Multiple test design techniques (Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition) have been applied.

5 Conclusion

The test case design provides comprehensive verification of the Smart Sports Stadium Operations Platform. Functional correctness, boundary behaviour, error handling, security enforcement, and key non-functional attributes are systematically validated. The combination of well-structured test cases, appropriate test design techniques, and clear requirements traceability ensures strong coverage and supports reliable delivery of a safety-sensitive stadium operations system.